



Capacity allocation games without an initial sequence



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ABSTRACT

We consider a capacity allocation problem where multiple manufacturers order production capacity. Orders generate a value and a time-dependent scheduling cost. No initial sequence of orders is assumed. Using a cooperative game, we investigate how manufacturers can order cooperatively to increase their profits. Assuming that each coalition expects known probabilities for its orders to be processed first and last, we identify exact conditions for the cooperative game to be balanced. We also show the intractability of the core emptiness test.

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1. Introduction

We consider a capacity allocation problem where a set of manufacturers orders production capacity from a common supplier. The capacity is time sensitive and each capacity unit corresponds to a time slot. The supplier has available time slots over a limited time horizon, which can be either sufficient or insufficient to process all the manufacturers' orders. Each processed order generates a value and incurs a scheduling cost for its owner. An unprocessed order has no value or cost for its owner. Order processing cannot be preempted. The manufacturers place their orders for capacity simultaneously and no initial sequence of their orders is assumed. We investigate how manufacturers can cooperate in making their ordering decisions to receive higher profit.

From a scheduling perspective, the supplier's capacity is the available processing time of a single machine and each order is a job with deterministic processing time. The related literature on sequencing games [4,5,3] and on subcontracting games with scheduling issues [1,2], mostly assumes a predetermined schedule, for example first-come first-served, of the orders. Then, the optimization problem faced by each set of manufacturers is to maximize the total value through resequencing/rescheduling over the initial schedule. Accordingly, a cooperative game can be naturally defined.

However, manufacturers may coordinate their ordering decisions prior to scheduling decisions, in which case there is no predetermined allocation of capacity to orders. Cooperative sequencing games without an initial order sequence have received limited attention in the literature. Maniquet [9] studies a similar capacity allocation problem, but assumes unit processing time of each order. He finds the Shapley value [13] of the cooperative game thus defined. Mishra and Rangarajan [10] allow for general order processing time, and characterize the Shapley value of the corresponding game. Klijn and Sánchez [8] prove the balancedness of the cooperative game defined over a similar capacity allocation problem where each set of orders is expected to be processed after all the other orders. Schulz and Uhan [12] consider the least core of a cooperative game defined over a similar capacity allocation problem where each set of manufacturers expect its orders to be processed before all the other orders.

Essentially, Klijn and Sánchez [8] assume highly pessimistic manufacturers, whereas Schulz and Uhan [12] assume highly optimistic manufacturers. The balancedness of the two resulting games is different: the game is balanced with pessimistic manufacturers and unbalanced with optimistic manufacturers. Interestingly, the Shapley values of the two games are the same [9,10]. We describe a more practical and reasonable alternative between these two extreme assumptions, where each set of manufacturers has a probability that its orders are processed first and last. We also identify exact conditions for the cooperative game to be balanced.

2. Capacity allocation game

We start by introducing relevant concepts in the theory of cooperative games. An n -player cooperative game (N, v) consists

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of a set $N = \{1, \dots, n\}$ of players, and a characteristic function $v : 2^N \rightarrow \mathbb{R}^+$ with $v(\emptyset) = 0$. A subset $S \subseteq N$ of the players is a coalition, and N represents the grand coalition. In the maximum value games we consider, the value $v(S)$ of the characteristic function represents the value that can be achieved by coalition S if all its members cooperate outside the grand coalition.

A central question within cooperative game theory is how to allocate the total value $v(N)$ of the grand coalition fairly among all the players. A standard approach is a core allocation [7]. If under a feasible payoff vector X that is a division of $v(N)$, i.e., $\sum_{i=1}^n x_i = v(N)$, no coalition can improve its total payoff by departing from the grand coalition, i.e., $\sum_{i \in S} x_i \geq v(S)$ for all $S \subset N$, then X is a core allocation. The set of all core allocations is the core of the game. A game with a nonempty core for all instances is *balanced*. These definitions follow conventions in both game theory and optimization (e.g., [11]). Specifically, a game is defined by mathematical representations, and an instance is a realization of a game with specific numerical values. Hence, balancedness is defined for a game, and core emptiness is defined for each instance.

A characteristic function is *convex* if, for any player i and two coalitions S_1 and S_2 satisfying $S_1 \subset S_2 \subseteq N/\{i\}$, we have $v(S_2 \cup \{i\}) - v(S_2) \geq v(S_1 \cup \{i\}) - v(S_1)$. Convexity is equivalent to supermodularity. A maximum value cooperative game with a convex characteristic function is a *convex game*. In a convex game, the marginal value of adding a player to each coalition increases as the coalition grows, and the game is balanced [13].

Consider a capacity allocation problem where a set of manufacturers, $N = \{1, \dots, n\}$, orders capacity from a common supplier. The supplier has a unit available capacity per time slot over a time horizon $1, \dots, T$. Each manufacturer i has an order to place, with capacity requirement i.e., order size p_i , value v_i if processed, and operational cost $w_i C_i$ if completed at time C_i , where w_i is a non-negative constant referred to as *order weight*. We assume that each order is profitable even if it completes at time T , i.e., $v_i \geq w_i T$. Preemption of order processing is not allowed. Let P denote the total capacity requirement of the n manufacturers, i.e., $P = \sum_{i=1}^n p_i$. Since a customer has no incentive to order any capacity at a time slot later than P , the case with sufficient capacity can be modeled as $T = P$. However, in the classical scheduling literature, it is common to consider the case of insufficient capacity, here defined by $T < P$. This case occurs frequently in practice, and therefore requires investigation as part of our work.

We further assume that every order is either processed completely or not at all. We consider a manufacturer, or equivalently an order, as a player. From the perspective of core allocation, assuming that each manufacturer has a single order is not restrictive, since a manufacturer having multiple orders can be viewed as a coalition of multiple manufacturers each having a single order. Let $p_S = \sum_{i \in S} p_i$ denote the total capacity required by coalition S .

We define a *head–tail allocation (HTA)* characteristic function $v(S)$ for the capacity allocation game:

$$v(S) = q\mathcal{A}(S, R_{S,h}) + (1-q)\mathcal{A}(S, R_{S,t}), \quad (1)$$

where $R_{S,h}$ is the set of time slots $1, \dots, \min\{p_S, T\}$ in a *head allocation* allocated with a *head probability* q , $R_{S,t}$ is the set of time slots $P - p_S + 1, \dots, T$ in a *tail allocation* allocated with a *tail probability* $1 - q$, and for $k \in \{h, t\}$, $\mathcal{A}(S, R_{S,k})$ is the value of the solution found by an Algorithm $\mathcal{A}$ for coalition S using time slots in $R_{S,k}$.

The head probability is a measure of the degree of optimism of each coalition: a more optimistic coalition would assume a higher value of the head probability. The consideration of head and tail allocations, with total probability of 1, has a useful economic interpretation: each firm incorporates best case (head allocation) and worst case (tail allocation) scenarios into its

decision making. This approach provides a convenient way for firms competing for time sensitive capacity to rationalize their expectations when cooperating over their decisions. The head–tail allocation generalizes the consideration of head-only and tail-only allocations in the literature [9,10,8,12].

We consider the capacity allocation game with sufficient capacity to process all orders, i.e., $T = P$. Smith [14] shows that the *shortest weighted processing time (SWPT)* rule, where orders are sequenced by nondecreasing ratio of size to weight, minimizes the total weighted completion time in single machine processing. Therefore, we assume that Algorithm $\mathcal{A}$ in the characteristic function (1) uses the SWPT rule and thus is optimal. We show that under a weak condition on q , the HTA value defines a convex game for the capacity allocation problem.

Theorem 1. *For the capacity allocation game where the capacity is sufficient and each order has a linear profit, the HTA defines a convex and thus balanced game if $q \leq 0.5$. Moreover, the condition $q \leq 0.5$ is necessary.*

Proof. Consider a player i and two coalitions S_1 and S_2 , where $S_1 \subset S_2 \subseteq N/\{i\}$. Assume that orders are indexed by the SWPT rule. Let S_1^A and S_2^A be the sets of orders within S_1 and S_2 , respectively, where the index of any order in S_1 or S_2 is larger than i . Let $S_1^B = S_1 \setminus S_1^A$ and $S_2^B = S_2 \setminus S_2^A$. Let $w_S = \sum_{i \in S} w_i$ be the total weight of orders in $S \subseteq N$. We have

$$\begin{aligned} & [v(S_2 \cup \{i\}) - v(S_2)] - [v(S_1 \cup \{i\}) - v(S_1)] \\ &= q(v_i - w_i(p_{S_2^B} + p_i) - w_{S_2^A} p_i) \\ & \quad + (1-q)(v_i - w_i(P - p_{S_2^A}) + w_{S_2^B} p_i) \\ & \quad - q(v_i - w_i(p_{S_1^B} + p_i) - w_{S_1^A} p_i) \\ & \quad - (1-q)(v_i - w_i(P - p_{S_1^A}) + w_{S_1^B} p_i) \\ &= (1-q)p_i(w_{S_2^B} - w_{S_1^B}) - qw_i(p_{S_2^B} - p_{S_1^B}) \\ & \quad + (1-q)w_i(p_{S_2^A} - p_{S_1^A}) - qp_i(w_{S_2^A} - w_{S_1^A}) \\ &\geq [p_i(w_{S_2^B} - w_{S_1^B}) - w_i(p_{S_2^B} - p_{S_1^B}) \\ & \quad + w_i(p_{S_2^A} - p_{S_1^A}) - p_i(w_{S_2^A} - w_{S_1^A})]/2 \\ &= [(p_i w_{S_2^B \setminus S_1^B} - w_i p_{S_2^B \setminus S_1^B}) + (w_i p_{S_2^A \setminus S_1^A} - p_i w_{S_2^A \setminus S_1^A})]/2 \\ &\geq 0, \end{aligned}$$

where the first inequality follows from the assumption that $q \leq 0.5$ and $S_1 \subset S_2$, and the second inequality holds since (a) $p_i/w_i \geq p_j/w_j$ for any $j \in S_2^B \setminus S_1^B$ if j exists, and thus $p_i/w_i \geq p_{S_2^B \setminus S_1^B}/w_{S_2^B \setminus S_1^B}$, and (b) $p_i/w_i \leq p_j/w_j$ for any $j \in S_2^A \setminus S_1^A$ if j exists, and thus $p_i/w_i \leq p_{S_2^A \setminus S_1^A}/w_{S_2^A \setminus S_1^A}$. Therefore, the game is convex and thus balanced.

We now show that the condition $q \leq 0.5$ is necessary for balancedness. Consider an instance with $n = 3$, $p_1 = p_2 = p_3 = 1$, $v_1 = v_2 = v_3 = 3$, and $w_1 = w_2 = w_3 = 1$. Let $q = 0.5 + \delta$, where $0 < \delta \leq 0.5$. Now, $v(N) = 3$, and a coalition of any two orders has value $2 + 2\delta$. If we let (x_1, x_2, x_3) denote a payoff vector in the core, we must have $x_1 + x_2 \geq 2 + 2\delta$, $x_1 + x_3 \geq 2 + 2\delta$ and $x_2 + x_3 \geq 2 + 2\delta$, i.e., $x_1 + x_2 + x_3 \geq 3(2 + 2\delta)/2 = 3 + 3\delta > 3 = v(N)$, a contradiction. That is, for any $q > 0.5$, there exists a game instance with empty core, and thus the game is unbalanced. $\square$

When capacity is insufficient, i.e., $T < P$, condition $q \leq 0.5$ does not guarantee balancedness of the game. This is shown by the instance constructed in the proof of Theorem 2, which shows the intractability of deciding whether a given game instance has a nonempty core. We note that when $T < P$, even computing the value of the grand coalition is NP-hard. However, this result does not imply intractability of the core emptiness test. The proof of Theorem 2 is included to make the paper self-contained.

Theorem 2. *The problem of deciding whether an instance of the capacity allocation game with $T \leq P$ has an empty core is binary NP-hard for any $0 < q \leq 0.5$, even when the weight of each job is 0.*

Proof. By reduction from the following problem.

Partition [6]: Given $2t$ elements with positive integer sizes $a_1, \dots, a_{2t}$, where $a_i < A$ for $i \in \{1, \dots, 2t\}$, and $\sum_{i=1}^{2t} a_i = 2A$, does there exist a partition S_1, S_2 of the index set $\{1, \dots, 2t\}$ such that $\sum_{i \in S_1} a_i = \sum_{i \in S_2} a_i = A$?

Given an instance of Partition, we construct an instance of the capacity allocation game, for each of the following four cases.

Case 1: $0 < q < 1/3$. Let K be the largest integer no greater than $1/q - 2$. If $1/q - 2 = K$, then let $n = 2t + K$, $T = A$, $v_i = p_i = a_i$ for $i = 1, \dots, 2t - 1$, $v_{2t} = p_{2t} = a_{2t} + K\varepsilon$, $v_i = p_i = A - \varepsilon$ for $i = 2t + 1, \dots, 2t + K$, and $w_i = 0$ for $i = 1, \dots, 2t + K$, where $0 < \varepsilon < 1/K$ is a constant. If $1/q - 2 > K$, then let $n = 2t + K + 2$, $T = A$, $v_i = p_i = a_i$ for $i = 1, \dots, 2t - 1$, $v_{2t} = p_{2t} = a_{2t} + K\varepsilon$, $v_{2t+1} = 0$, $p_{2t+1} = \varepsilon$, $v_{2t+2} = p_{2t+2} = ((1/q - 2) - K)A - \varepsilon$, $v_i = p_i = A - \varepsilon$ for $i = 2t + 3, \dots, 2t + K + 2$, and $w_i = 0$ for $i = 1, \dots, 2t + K + 2$, where $0 < \varepsilon < \min\{1/K, ((1/q - 2) - K)A\}$ is a constant.

Case 2: $q = 1/3$. Let $n = 2t + 1$, $T = A$, $v_i = p_i = a_i$ for $i = 1, \dots, 2t - 1$, $v_{2t} = p_{2t} = a_{2t} + \varepsilon$, $v_{2t+1} = p_{2t+1} = A - \varepsilon$, and $w_i = 0$ for $i = 1, \dots, 2t + 1$, where $0 < \varepsilon < 1$ is a constant.

Case 3: $1/3 < q < 1/2$. Let $n = 2t + 2$, $T = A$, $v_i = p_i = a_i$ for $i = 1, \dots, 2t$, $v_{2t+1} = 0$, $p_{2t+1} = \varepsilon$, $v_{2t+2} = p_{2t+2} = (1/q - 2)A - \varepsilon$, and $w_i = 0$ for $i = 1, \dots, 2t + 2$, where $0 < \varepsilon < 1$ is a constant.

Case 4: $q = 1/2$. Let $n = 2t$, $T = A$, and for each order i , $v_i = p_i = a_i$ and $w_i = 0$.

It is straightforward to show, in each of the four cases above, the instance of the capacity allocation game has a nonempty core if and only if the instance of Partition has a solution. $\square$

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